

Abstract:

For millions across the globe, the cultivation of illicit drug crops, such as coca bush (the primary ingredient used in the manufacture of cocaine), continue to represent an important, but often precarious, part of their broader livelihoods. In Peru, currently the largest supplier of the crop, dependence on coca has surged to record levels in recent years - from 54,635 hectares (ha) under cultivation in 2019 to more than 90,000 ha in 2023, according the country's drug control agency, DEVIDA (DEVIDA 2024b). The vast majority, however, is grown by communities of rural, peasant farmers, henceforth 'illicit peasantries', concentrated within the country's Amazonian territories. Dominant explanations for the high levels of engagement with illicit drug crops amongst individuals and households in so-called 'frontier spaces' emphasise the interplay of 'state absence' and 'deficits of development'. This project seeks to better understand how state drug control regimes shape livelihood outcomes in 'narco-frontiers', such as the Peruvian Amazon.

We do so by approaching systems of territorial governance from a critical, 'off-centred lens', as the manifestation of diverse interests that are held and bargained for by both 'the governings' and the 'governed'. In this way, we hope to more effectively understand where drug control strategies have succeeded and failed, and to (re)interrogate what that means, and for whom. In so doing, we also attempt to contribute to a significant empirical gap in the current literature. As of writing, very little empirical analysis has been directed at understanding Peru's drug control regime during the latest period of sustained coca expansion, with the exception of a limited number of individual ethnographic case studies. Even less has been conducted on the Amazon from a regional, 'narco-frontier' perspective. By leveraging novel sources of remote sensing data, capable of localising individual territories of cultivation, alongside a robust, linear dynamic panel model, designed to capture both the persistence of coca livelihoods and the 'erraticism' of Peru's current drug control regime, we offer some carefully placed critiques of current policy frameworks, particularly its reliance on coercive, forced eradication strategies. In so doing, we hope to make a modest contribution to a growing corpus of work dedicated to drug crop growers' perspectives, which are too often fetishized by dominant discourses in the Policy Arena, and too scarcely made the subject of serious study.

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Glossary of Terms and Acronyms

AD	Alternative Development
AL	Alternative Livelihoods
CONPACCP	The Confederación Nacional de Productores Agropecuarios de las Cuencas Cocaleras del Perú
CORAH	Control Y Reducción Del Cultivo De La Coca En El Alto Huallaga <i>English translation:</i> Special Project of Control of Coca Crops in the Upper Huallaga
DEVIDA	Comisión Nacional para el Desarrollo y Vida Sin Drogas <i>English translation:</i> The National Commission for Development and Life without Drugs
ENACO	Empresa Nacional de la Coca <i>English translation:</i> The National Enterprise of Coca
IDC	Illicit Drug Crops
IDE	Illicit Drug Economies
PIRIDAIS	El Programa Presupuestal de Desarrollo Alternativo Integral y Sostenible <i>English translation:</i> Comprehensive and Sustainable Alternative Development Budget Program
RAISG	Rede Amazônica de Informação Socioambiental Georreferenciada <i>English translation:</i> Amazon Network of Georeferenced Socio-Environmental Information

1. Introduction

In Peru, dependence on coca has surged to record levels in recent years - from an estimated 54,635 hectares (ha) under cultivation in 2019 to more than 90,000 ha in 2023 (DEVIDA 2024b). The vast majority is grown by communities of rural, peasant farmers, or 'illicit peasantries' (Goodhand, Ballvé, et al. 2024), concentrated within the country's Amazonian territories (UNODC 2023b, tbl. 1).

Against this backdrop, dominant 'development-oriented' explanations for households' engagement with illicit drug crops in so-called 'frontier spaces', and the role played by the state and drug control policy in shaping those livelihood trajectories, have come under significant criticism. This project, therefore, seeks to address the following question: *How do state drug control regimes shape livelihood outcomes in 'narco-frontiers', such as the Peruvian Amazon?*

We focus on the state, given its undeniable importance in mobilising the resources necessary to address significant problems of development at scale, but understand it more critically, through an 'off-centered' lens (Krupa and Nugent 2015) that sees its position and influence as both determined and constrained by relational dynamics between governing elite interests and 'the governed' drug crop growers.

On these bases, drawing from the literature on territorialisation, political settlements and the broader critical development literature, we propose a synthesis, 'socio-territorial' approach, that is broadly consistent with that advocated for by a growing number of Peruvian and Andean researchers (Paredes and Pastor 2024; Svampa 2019; Yáñez et al. 2021; Gonzalo Delgado et al. 2024) but has remained somewhat confined to these local discursive contexts. As the basis for an empirical study, some promising initial findings from novel sources of remote sensing data provide merit for more serious consideration of these local perspectives, both within the Western-dominated drug control policy arena (Buxton 2015) and the (similarly Western-dominated) landscape of academic research.

2. Background

2.1 A Brief History: Coca Crop Growers and the Peruvian State

Whilst Peru's illicit economy encapsulates several illicit drug crop markets, including an established marijuana supply chain, coca (*Erythroxylum coca*) is by far the most extensive (UNODC 2022; DEVIDA 2024b), with an estimated 99% of that cultivation understood to be located within the Amazon basin (UNODC 2023b, 66). Drug crop-growers, or 'illicit peasantries', cannot be said to represent one homogenous group (Goodhand, Ballvé, et al. 2024), however, the majority within the region are smallholder growers, defined by DEVIDA as owning farms smaller than 5 hectares¹ (DEVIDA and PCM 2019, 7).

In addition, a significant number of coca growers in the Peruvian Amazon participate through less formal means, such as day-labour coca pickers, or *raspachín*, many of whom migrate from neighbouring areas (Silverstein 2021). In many cases, coca also does not represent the entirety of any singular livelihood (Grillo et al. 2021). Instead, it often represents one, albeit important, livelihood strategy that is employed alongside others, either to diversify risk (as in the case of intercropping), see out broader economic shocks, or meet other budgetary shortfalls in the household. Large-scale monocropping of coca has typically been the exclusive domain of a much smaller number of organised criminal groups or other local elites, who are sufficiently resourced to address the greater risk of detection and intervention by law enforcement agencies posed (either through leveraging of local connections, bribery, or through violence) (E. Dávalos and Dávalos 2019; Goodhand, Meehan, et al. 2024).

Rising cultivation levels have, in a loose sense, tracked a broader increase in global demand, particularly in North American and European markets (UNODC 2023a). As a result of pronounced expansion within the country, however, Peru has recently become recognised as the single largest grower of the crop in the world, overtaking Colombia and Bolivia (UNODC 2023a). Scholars like (Gootenberg 2017) have illustrated that this relatively recent picture of the country, from the perspective of the prohibitionist international drug control regime, in many ways sits in tension with Peru's long and changeable relationship with coca.

¹ Recent official estimates suggest that 53% of plots are less than 4 hectares, with 82% less than 10 hectares (INEI 2015). Other estimates suggest a majority produced on those between 0.5 and 2 hectares (Grillo 2018)

Peru has long been the site of traditional coca use among Andean peoples. Dating back to the Inca Empire, indigenous communities valued coca for its stimulative effects and relief from altitude sickness and hunger, either by chewing the leaves or brewing them for tea (Grisaffi 2021). In spite of this history, Gootenberg argues that much of Peru's public consciousness has experienced 'collective amnesia' regarding its relationship with the crop (Gootenberg 2017, 28). This is perhaps most apparent in the period beginning in the 1880s through to the 1940s, when Peru saw intense legal cocaine production. The state and economy became closely tied to the market, with Huanaco rising as a global exporter of pharmaceutical cocaine, led by economic elites, including "doctors, statesmen and capitalists" (Paul Gootenberg 2001, 6), and supported in significant part by the state (Paul Gootenberg 2001).

It was only during the mid-20th century that cocaine, and coca by association, gained its current harmful reputation, as the "pariah drug". This process was largely initiated by the U.S., which had increasingly begun to characterise the trade as a national security threat, which was formalised by the UN Commission of Enquiry on the Coca Leaf to Peru and Bolivia in 1948-50 (Gootenberg 2017, 28). Under threats to withdraw (or make conditional) sources of development aid, Peru not only underwent a rapid shift away from its previous position on coca and cocaine, but also a systematic revision of its historical relationship to them (Gootenberg 2017)²

2.2 The Evolution of Drug Control Policy

This changeability has, in many ways, been reflected by the experimentation and "Mixed Messages" of Peruvian drug control policy (AIN 2020). A recent report by the Andean Information Network described the state's historical record as "erratic, shifting between negotiated gradual reduction with alternative development to forced eradication." (Grisaffi et al. 2020).

During what might be called the peak of Peru's 'drug war' policy agenda, from the mid-20th to early 21st century, Peru's national drug control policy agenda almost entirely prioritised traditional, counternarcotics strategies, including law enforcement patrols, militarised interdiction operations, and forced eradication campaigns (Buxton 2015). Until around the 1990s, the latter, in particular, represented the greatest focus of national drug control policy (Garcia-Yi 2014).

² See further on history of settler-colonial expansion in the Amazon as a driver of cocaine markets (L. M. Dávalos et al. 2016; Paredes and Manrique 2021; Gootenberg and Dávalos 2018)

Amongst the most significant of these policy programmes has been the ‘CORAH’ programme, established in 1982, under the Peruvian Ministry of the Interior. Whilst originally intended to focus its operations in the Upper Huallaga valley, CORAH’s mandate has since been broadened to encompass all coca-growing territories in Peru, and now represents the primary policy channel through which forced eradication campaigns are implemented (DEVIDA 2023b). These campaigns are carried out by teams of specialised eradication agents who manually remove coca plants and seedlings in identified areas of activity, typically through the use of a mechanical implement, or *cococho* (DEVIDA 2023b, 144), and in accompaniment with members of law enforcement. These programmes have aroused strong opposition from coca-growing communities, often culminating in violent clashes and even deaths (Grisaffi et al. 2020; Garcia-Yi 2014).

A notable exception is the country's limited experimentation with legalisation that began in 1949, with ‘Estanco de la Coca’, and was continued through its successor, ENACO in 1978 (Busnel and Manrique 2023). In part, to recognise the country’s heritage, but substantially in response to growing upheaval from coca union and peasant protests during the 1960s (Gootenberg and Dávalos 2018, 5), the state officially established a monopoly buyer of legal coca, alongside a register of legal growers (Busnel and Manrique 2023, 3). Whilst important in understanding Peru’s broader evolution and vacillation in policy vision, ENACO represents no more than 2% of the overall coca supply, lacking the comparable scale of other policies currently in effect and is not a focus of subsequent discussion.

Next followed several short-lived variations of ‘Alternative Development’, ultimately resulting in the adoption of the current PIRDAIS programme in 2014. PIRDAIS, at some level, marks progress towards a more comprehensive development-focused drug policy, targeting three broad objectives: building licit agricultural supply chains, improving state-society relations in coca regions, and reducing rural families’ involvement in illicit economies by enhancing local conditions (DEVIDA and PCM 2019; Paredes et al. 2024, 32). Interventions include more traditional crop substitution programmes (like plantain, cacao, and coffee), better rural roads, and broader governance initiatives, focusing on coca-growing or at-risk areas (DEVIDA and PCM 2019). Yet, PIRDAIS has not been a fully distinct phase. Critics note overlaps with forced eradication, leading to “a cycle of unfulfilled promises that undermine the legitimacy of the state in the territory” (Paredes et al. 2024, 56) - See more detailed table in the Appendix.

3. Conceptual Frameworks

3.1 Positivist Policy Logics in ‘Big D Development’

Considering the widespread failures of the War on Drugs³ a growing consensus among policymakers has been the need for reform within the drug control agenda. Increasingly, this has involved appeals to ‘development-oriented approaches’ in mainstream discourse, as a means of providing a ‘solution’ to illicit drug economies (IDEs) that more closely align with the state’s existing commitments to development and peacekeeping-related objectives, such as the Sustainable Development Goals (SDGs) (Goodhand et al. 2021). Development, here, is considered the projects of “*statebuilding, economic growth, poverty reduction, and reducing levels of armed violence.*” (Goodhand et al. 2020, 13).⁴

Many of the approaches currently pursued in major drug producing regions, such as Afghanistan, Myanmar and Latin America (particularly Bolivia, Colombia and Peru), may be characterised by a ‘big D’, Washington Consensus-style of Development (Mansfield 2016; Goodhand, Ballvé, et al. 2024)⁵, in which increasing productive output is seen as the primary engine of progress, unlocked through integration into liberal markets and democratic institutions of governance (Goodhand et al. 2021).

‘Alternative Development’ (AD), defined by the United Nations Office on Drugs and Crime, places the objective of “sustained national growth” at the forefront:

“a process to prevent and eliminate the illicit cultivation of plants containing narcotic drugs and psychotropic substances through specifically designed rural development measures in the context of sustained national growth and sustainable development efforts in countries taking action against drugs” (UNGA 1998).

Within this particular framing of Development, IDCs and IDEs have typically been framed “as residual to, or undermining of” the necessary processes (Goodhand et al. 2020, 13): as,

3 Criticism includes not only a lack of long-term reductions in the supply of illicit drugs (Shaw 2016), but also the “systematic human rights onslaught” that such approaches have been accused of in many affected regions (Barrett 2014; Buxton 2015).

4 Except for the latter, which does not provide a central focus of the current study, it is these dimensions of development that we will attempt to return to throughout.

5 See the work of (Mawdsley and Taggart 2022) for a nuanced analysis of the ‘big D-little d’ distinction in development policy and literature, including the historic roots of both.

everywhere and always, a symptom and a barrier to development. ‘Deficits of development’ (Brombacher and Westerbarkei 2019), such as lack of access to productive resources, high rates of poverty and a lack of integration with established (licit) markets, leave drug crop producers without competing legal livelihood strategies (UNODC 2023b, 73; 2017). Economic challenges of this nature are regularly observed amongst remote and marginalised drug crop-growing households in ‘frontier’ communities, described by one report as “[an] environment where poverty, exclusion and marginalization are the norm, and where informal and often illicit economic activities tend to become a means of subsistence.” (Amazon Underworld 2023, 6).

Present discourses regularly also highlight the role that IDCs play in blocking attempts to improve conditions by stalling those same processes of development, as part of a ‘vicious cycle’. At the level of growers, reports describe “the poverty trap of illicit crop cultivation” (UNODC 2017), citing drug crop markets’ volatile wages and exploitative working conditions as unsuitable for providing the conditions to help growers out of poverty in the long-term (Grisaffi et al. 2020; Goodhand 2005; Mansfield 2013). State actors also highlight how IDCs undermine broader ‘pro-development’ processes in contexts of drug crop cultivation, such as institutions of ‘good governance’ and sources of private capital investment (Zeiler et al. 2019).

Discourses commonly highlight the destabilising potential of ‘bad actors’, such as the FARC paramilitary organisation in Colombia and the Taliban, previously, in Afghanistan (Mansfield 2017), who have relied upon revenues from IDEs to maintain their presence in drug crop-growing territories (Williams and Felbab-Brown 2012; Díaz and Sánchez 2004). This has made it more difficult for the state to (re)assert control over drug-producing territories to provide the stability and predictability that is required by businesses and owners of private capital to invest in the long-term growth of local, legal economies (Thomson et al. 2024). This is to say little of the wider ‘social ills’ related to ‘spillover violence’ in neighbouring territories, rising instances of drug (ab)use amongst domestic populations (UNODC 2023b; Shaw 2016; International Crisis Group 2024) and other challenges that have been put to incumbent governments.

From a development perspective, IDCs have, in many cases, therefore become a problem that both can, and should, be ‘solved’ by the state, often through implementation of one or several ‘win-win’ solutions. This is classically accomplished through a discourse of neutrality and the implementation of technical policies (Paredes and Manrique 2021), directed towards the

realisation of a more productive and, therefore, more prosperous citizenry (Goodhand 2021). In turn, coca crops become constructed as the less-preferred choices of desperate households, who would “automatically prefer” (Gutiérrez-Sanín et al. 2024, 1006) a legal alternative source of livelihood if it were made available to them.

3.2 Policy Frameworks and Critical Perspectives

‘Carrot and Stick’ frameworks have been at the forefront of many current development-oriented approaches (Rojas Herrera and Dessein 2023), seeing the institutions of governance as mechanisms through which to (re)align incentives with particular patterns of economic activity. Namely, to reduce the incentives associated with drug crop livelihoods (to provide the ‘sticks’) and increase the incentives associated with legal alternatives (the metaphorical ‘carrots’). Generally speaking, ‘Carrot and stick’ frameworks have tended to model producers’ behaviour from a Beckerian, ‘economics of crime’ approach⁶, in which households are modelled as (im)perfectly rational economic agents, whose choice of livelihood strategy largely reflects the price signals and expected payoffs associated with them (Becker 1968; E. Dávalos 2017). All else being equal, where high-value alternative sources of income are sparse in growing regions, the significant premium that coca leaves can earn at the farm gate (the so-called farmgate price) therefore creates a substantial incentive for producers to engage in those markets.

Beyond crop prices, households weigh the overall ‘net’ benefit of production, factoring in costs such as fertiliser, land rent, or crop loss (V. A. Greenfield et al. 2017; Mansfield 2013) and the ‘disutility’ of punishment, tied to its severity and likelihood of detection. Households pursue the livelihood option with the highest ‘net utility’ among available choices (Andersson 2013).

‘Sticks’ may be mobilised throughout the supply chain. In Peru, this includes stricter sentencing for farmers, manufacturers, and traffickers identified by law enforcement (see **Figure 1**) (E. Dávalos and Dávalos 2019), or increased forced eradication funding under CORAH (PCM 2022) to reduce profits at the cultivation stage, by increasing the risk of lost harvests (Buxton 2015; Dion and Russler 2008; Meehan and Dan 2023; Mansfield 2020).

⁶ In reference to the work of Gary Becker, best-known for presenting this approach in his seminal text: ‘Crime and Punishment: An Economic Approach’ (Becker 1968)

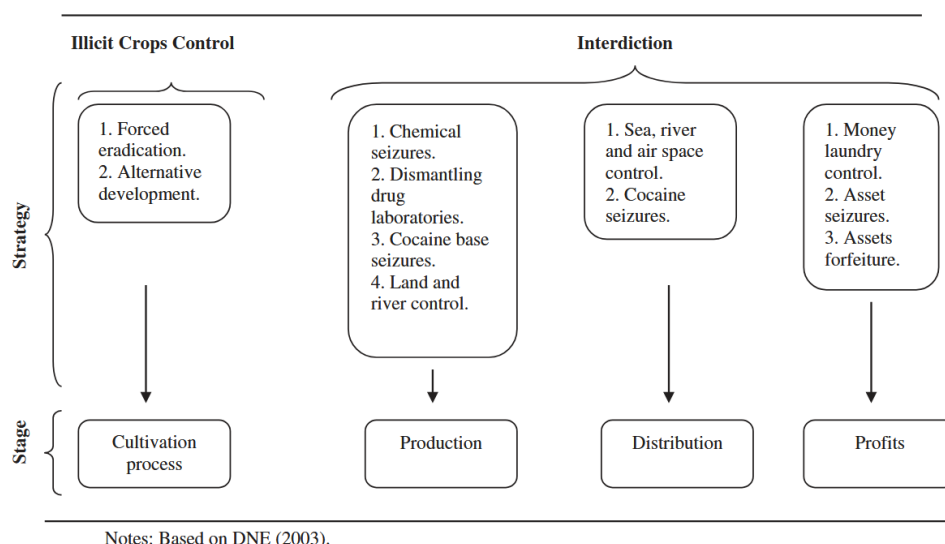


Figure 1: Policy ‘Sticks’ in the Economics of Crime Approach in Colombia
(E. Dávalos and Dávalos 2019, app. 1)

By contrast, ‘carrots’, more recently under AD initiatives, have come in the form of crop subsidy programmes, intended to supplement reduced profit margins in the legal market, physical and human capital investment, to provide farmers with the technical equipment and skills, as well as broader interventions designed to improve access to established national markets (Paredes et al. 2024; PCM 2025).

Scholars have given several reasons to be wary of these lines of thinking, however. For one, the explanatory power of such models, to reliably predict households’ livelihood responses based on positive or negative shocks to material ‘payoffs’, has been limited at best. Contradictory evidence from numerous large-scale crop substitution programmes that have failed to induce long-term transitions to alternative crops (Mansfield 2017; Farthing and Grisaffi 2023; Acero and Machuca 2021), enforcement-led interventions that have either steeled drug crop growers’ resolve (Paredes et al. 2024) or merely displaced pockets of activity to a neighbouring region (à la the ‘balloon effect’ (López et al. 2016)) have each suggested that households’ decision-making cannot be reduced to a simple economic calculus.

3.3 Theoretical Frameworks

While acknowledging the development challenges of prolonged dependence on IDCs, a ‘people-centred’ framework (United Nations 2024) must also consider the harm from failing to provide viable livelihood alternatives in their place. Grillo (Grillo 2018) notes that coca-growers in Peru seek not just income, but also resilience to shocks, access to services like

healthcare and education, and the capacity to meet basic needs; drug crop markets often also compensate for gaps left by absent or unreliable state services (Goodhand et al. 2020).

Greater attention must be paid to how the state and its policies affect governed populations, including perceptions of authority and trust among drug crop growers (Acero and Thomson 2022; Paredes et al. 2024). This involves questioning the ‘neutral central agent’ model of statebuilding and recognising that “*the state does not manifest itself in a uniform way to all groups*” (Jawad et al. 2021, 6). Variations in authority and legitimacy, shaped by state presence and service delivery, help explain the ‘territorial heterogeneity’ seen in responses to Peru’s drug control agenda (Paredes et al. 2024; Krupa and Nugent 2015; Silverstein 2021).

In conversation with this, Paredes and colleagues within the Centre for Rural Development in Latin America (RIMISP) research group, describe a much-needed ‘socio-territorial approach’ (Paredes et al. 2024).

"[socio-territorial conflicts] CSTs are complex social processes marked by power asymmetries, in which diverse interests of collective actors (territorial and extraterritorial) are manifested, as evidence of the different ways of understanding territory and its development. [...] we analyse tensions, disputes, and/or controversies that affect territories regarding their functioning and governance, access to and use of their resources, how these are distributed, and the criteria that define the inclusion/exclusion of different social actors in the deliberation and decision-making that determine their development trajectories" (RIMISP 2023)

The extent to which particular systems of resource rights, or territorial governance regimes, privilege particular groups or serve certain livelihood outcomes is a question of the social contract that emerges between those claiming authority over a space and people (those attempting to govern a territory, such as the state), and those being governed (Rasmussen and Lund 2018; Corson 2011). Territorialisation, as it is broadly defined by Rasmussen and Lund, refers to “*a strategy of using bounded spaces for particular outcomes ... [a resource control strategy] that involves the classification of particular areas to regulate people and resources.*” (Rasmussen and Lund 2018, 388). They amount to a claim over the economic resources that can be accessed by specific groups, as well as how they may be used, and are therefore intimately related to how livelihoods and livelihood outcomes are regulated.

Paul Nugent has evaluated these in terms of ‘inclusive’, ‘productive’, and ‘coercive’ dynamics of control between the ruling party and the ‘ruled’, corresponding to varying extents of the latter’s interests being accounted for in the formers’ system of territorial rule (Nugent 2010, 43–44): coercive contracts being the most unaligned with those interests, and inclusive being the most aligned. In this project, we explore not just when the state’s drug control policies lead to a reduction in drug crop cultivation, but also when those reductions are achieved as part of a transition to alternative livelihoods that reflect these valued livelihood outcomes.

However, authors have stated there is no guarantee of ‘the state’ accepting the interests of drug crop growers as part of an inclusive social contract (Putzel and Di John 2009). More or less inclusive social contracts are determined by the set of interests that are held by relevant groups and persons, and their respective abilities to have those interests recognised through a process of bargaining (Putzel and Di John 2009). Contrary to the suggestions of much ‘state realist’ discourse, including those commonly associated with liberal developmentalist statebuilding concepts (Nugent 2010), political settlement theorists argue that discussion of ‘states’ and ‘state interests’ in this context can prove unhelpful. Both concepts have referred to significantly different phenomena across different territorial contexts and times, and refer to no fixed ‘thing’ (Putzel and Di John 2009).

This is particularly clear in the Peruvian context, where the state has alternated between outright support for coca-cocaine production (Gootenberg 2017; Paul Gootenberg 2014) and disengagement, especially in more remote regions. As several note, only when prominent competing interests emerged, in the form of private capital and US foreign policy, did the state adopt more forceful, anti-grower narratives: first as ‘criminals’ during the ‘War on Drugs’, now as “victims of underdevelopment and obstacles to development progress” (Lu et al. 2022, 3; Silverstein 2021). Consistent with political settlements theory, these shifts in modes of territorial control may be explained with respect to the underlying interests of the political elites that have collectively determined where and when ‘the state’ has decided to exercise control (Gutiérrez-Sanín et al. 2024).

‘Extractive frontiers’ like the Amazon present diverse set of interests to consider within this framework in the form of ‘extractive imperatives’, relating to the economic rents that may be earned through appropriation of natural resources such as land, oil, lumber or valuable minerals

(Snyder 2006). Catherine Boone's (Boone 2003) model of extractive imperatives offers a promising explanation of both the role of these competing interests in the bargaining process, and of the historic 'unevenness' of state interventions and (re)assertions of control in narco-frontiers: Where extractive imperatives exist in a territory, 'active engagement' from local elites, to gain favourable access to those rents, may be expected. Otherwise, a territory, including its people, resources and existing livelihoods, may be left to its own devices (Boone 2003). When those interests "run up against the reality of cohesive local interests", as Nugent recounts, or those interests of other local elites, "some form of negotiation is likely to ensue." (Nugent 2010, 40).

This does not mean 'the state' is insignificant in shaping policy outcomes. Unlike top-down 'Carrot and Stick' models (E. Dávalos and Dávalos 2019), or even more recent political settlement analyses (Goodhand and Mansfield 2010) which overlook relational dynamics, we argue that recognising how 'the state' is constructed in the minds and experiences of growers refines theoretical frameworks, especially for understanding responses to development-oriented control strategies.

By acknowledging that material incentives, such as farmgate prices, play a necessary, but not sufficient role in explaining drug crop growers' choice of livelihood trajectories and by understanding the objectives of 'the state' as one of competing elite interests, particularly concerning extractive imperatives in extractive frontier territories, and finally, by allowing for variations in the way that "*the state is apprehended- how it is granted objective, known status, is "seen" and experienced*" (Krupa and Nugent 2015, 6) to mediate the outcomes of the policies it implements, we attempt to put forward several testable hypotheses in-line with this 'socio-territorial' theoretical approach.

3.4 Hypotheses Construction

Whilst our primary interest lies in understanding where and when attempts to reduce dependence on drug crop cultivation are achieved as part of an 'inclusive' settlement (Nugent 2010), referred to henceforth as an 'inclusive transition', there currently exist very few empirical studies attempting to address even the more basic underlying claim of where state policies have, or have not, induced reductions.

Where existing research designs have considered broader, national patterns of similarity and difference in the conditions and policies driving coca cultivation, such as potential frontier dynamics in the Amazon, they have tended to obscure important territorial level differences by aggregating at the regional-level or above (Balieiro and Nascimento 2014; López et al. 2016; Verdugo Yepes and Pedroni 2011).

In line with previous critiques of the harmful development outcomes realised by past policy approaches, it is helpful to first clearly illustrate the specific cases in which drug control policies have failed to realise an ‘inclusive transition’. For example, failing to meet certain conditions predicted by the present ‘socio-territorial’ framework.

This study will initially attempt to address the following research questions:

1. Under what conditions do coca-growing households move away from drug crop livelihoods?

Given the vastness of the Amazonian frontier and the role of extractive imperatives as we have conceived them (Boone 2003), we may expect that the state is more likely to ‘actively engage’ with drug crop cultivation, in terms of the efforts it expends to reduce cultivation, where doing so furthers pronounced local economic interests. This may include the removal of local coca crop farmers from spaces where their livelihoods are seen to undermine processes of capital accumulation (such as where their settlement on particular lands prevent access to unexplored resource deposits) or where the heightened risk profile, associated with areas of drug crop production, increases the borrowing costs imposed by larger-scale industries, such as those in agribusiness, looking to expand (Ballvé and McSweeney 2020; Goodhand, Ballvé, et al. 2024; Thomson et al. 2024).

Observations from elsewhere in the region would broadly suggest that the preferred approach of state policy would be that which allows those Development processes to resume expediently, and at the lowest possible cost (Ballvé 2019). A coercive strategy, consistent with advancing these interests, may be expected to produce an observable reduction in coca cultivation, at least locally.

H1a: Coercive drug control policies will be associated with significant reductions in coca crop cultivation.

More inclusive strategies of drug control, at least in the immediate term, may be expected to have a weaker negative relationship with levels of drug crop cultivation, given the significant lead-in times that can be required for households to adjust to alternative livelihoods (Grillo et al. 2021). Therefore:

H1b: Inclusive drug control strategies will have a weak negative relationship with coca crop cultivation, at least in the short term.

2. Under what conditions do households move away from drug crop livelihoods as part of an inclusive transition?

Given the tendency of coercive territorial regimes to neglect growers' interests, and therefore the reasons behind their initial engagement with drug crop cultivation, we may expect strategies of this type to have a negative, or at least no significant positive, relationship with realising their relevant Livelihood Outcomes:

H2a: Reductions in coca cultivation, associated with the introduction of Coercive drug control strategies, will have no significant relationship with any associated changes in Livelihood Outcomes.

By contrast, we suggest that the greatest prospect for drug crop growers' interests to be realised in the terms of a transition away from IDEs and into legal alternatives is when those terms can be effectively advocated for as part of a robust bargaining process with an actively engaged state:

H2b: The strongest corresponding increase in Livelihood Outcomes will be associated with reductions in coca cultivation that are associated with: the implementation of Inclusive drug control strategies; the presence of nearby extractive imperatives; and the existence of strong bargaining power on the part of coca growers.

4. Data and Empirical Strategy

To consider how particular state drug control regimes shape livelihood outcomes in the Peruvian Amazon, we will first set out an empirical design to capture some potential causal relationship between coercive or inclusive drug control strategies of a given character and some relevant measures of interest.

In the first part of our analysis, our outcomes of focus are the levels of coca crop cultivation observed in a given territory, the metric by which the state has typically evaluated the success of current drug control policy (PCM and DEVIDA 2021). We will then take a more critical perspective, considering the impact of these policies on livelihood outcomes that are more closely tied with established human development objectives, as they have been identified from previous research with drug crop-growing communities (Grillo 2018).

Subsequent discussion will explore any trends that are observed between the types of drug control regimes established and our different livelihood outcomes, referring to the potential socio-territorial dynamics set out in our previously established hypotheses. It will also reflect on any notable areas of divergence between these two different perspectives (coercive or inclusive), including in how they capture the role of the state in shaping recent livelihood trajectories in coca-growing regions.

See **Table of Descriptive Statistics** in Appendix for a list of variables and sources.

4.1 Data

By linking several traditional sources of administrative policy data with more novel sources of remote sensing data on specific agricultural territories, this project provides a more fine-grained analysis of more than 600 (N=612) coca-growing territories across the Peruvian Amazon during the period from 2019 to 2023. More specifically, we combine sources of district-level drug control policy data, including PIRDAIS spending levels and CORAH forced eradication target areas, with observed shifts in the area under coca cultivation, as reported by DEVIDA's official coca monitoring programme.

4.1.1 Dependent Variable: Drug Crops and Alternative livelihoods

Peru's coca monitoring programme, SISCOD builds upon previous collaborative work between the country's drug enforcement agency, DEVIDA, and the UNODC's Illicit Drug Crop Monitoring Programme (ICMP) (UNODC and DEVIDA 2018). This involved the combination of crop detection via satellite imagery, verified by expert analysis, and site visits (DEVIDA 2023b).

The use of satellite imagery and broader remote sensing techniques, to “measure and monitor the earth's surface by exploiting differences in the reflected or emitted electromagnetic radiation of surface objects” (Simms 2016, 4), has increasingly been recognised as a promising means for addressing some of the significant shortcomings of more traditional data collection methods in drug economies (Mansfield 2017). Unlike in-person field missions, which typically impose significant time and logistical costs, this method is particularly relevant in the context of more remote and challenging frontier geographies, such as the Amazon Basin, where such constraints have historically prevented official programmes from providing consistent and reliable data (Brombacher and Santos 2023; UNODC 2023b).

Like (E. Dávalos 2017, 53) we take our outcome of interest, Y , to be the difference in the area under coca cultivation, observed for some year period $t = 1$, relative to the previous year period $t - 1$.

$$Y_{it} = CocaCult_{it} - CocaCult_{it-1}$$

This choice of measure places focus on the dynamics of *change* in households' choice of coca cultivation as a livelihood strategy, rather than capturing trends in the overall levels of engagement, in absolute terms, in response to drug control strategies adopted by the state.

Insights from recent scholarship in drug crop-growing regions (Grillo 2018) have emphasised the need to also more closely ground understandings of ‘success’ and ‘progress’ in terms of local human development contexts, and the experiences of those most affected by state policies. That means going beyond seeing livelihoods as means to sustain ‘bare life’ (Bakker and Gill 2019), or trusting in the expansion of established legal markets to provide this. We leverage another source (MapBiomas 2025a) of remote sensing data to estimate activity in the local agricultural environments, such as livelihoods in Rice and other (licit) commodity crops that might be considered appropriate substitutes

4.1.3 Geographic and Administrative Data

Multiple competing definitions of the Amazon exist, both within the existing literature and across broader areas of discourse. ‘The Amazon’, as a geographic location, may be taken to refer to a wide range of concepts, including whether historical, hydrological, or legal lenses are considered (Eva and Huber 2005).

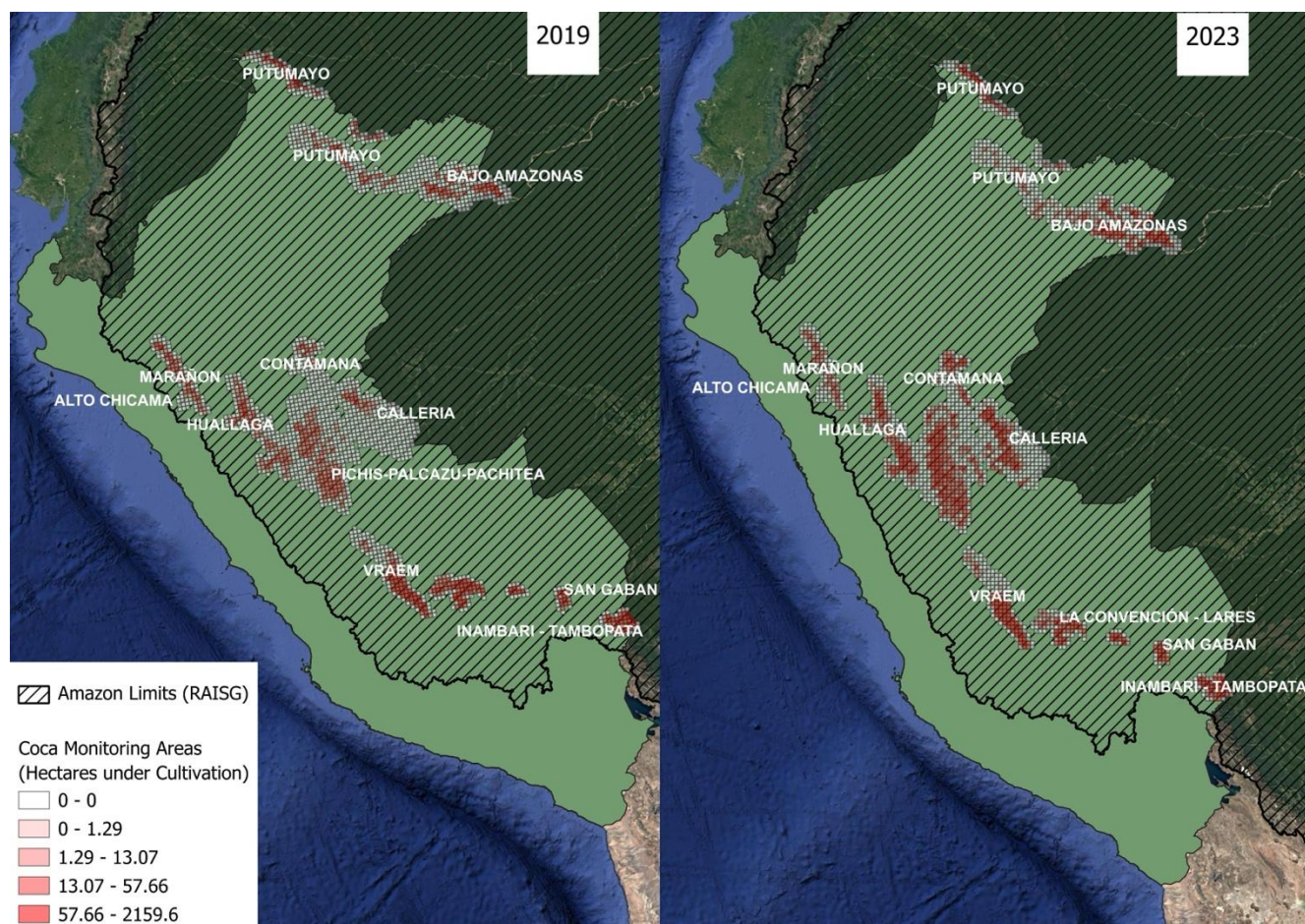


Figure 2: Coca Monitoring Areas in 2019 and 2023.

Constructed by the author using QGIS software version 3.40.6.

To analyse coca crop cultivation dynamics in the Amazon, as they relate to the domain of Peruvian state-led drug control policy, we adopt the ‘referential limits’ set out by RAISG⁷ (RAISG 2025). These boundaries not only provide a clear demarcation of the physical space occupied by the region, conducive to defining sample areas and conducting empirical observations, they also provide a means to usefully reflect the region’s politics, as they relate

⁷ RAISG is a globally recognised consortium of civil society organisations, representing across each of the Amazon countries, first formed to establish common terms of reference to facilitate productive and inclusive information exchange about the region.

to local understandings of territorial control and projections of power, without unnecessarily privileging the boundaries preferred by any one state over another.

Amazonia, thus defined, describes the entirety of the Amazon Basin⁸, in addition to a number of additional transitional spaces within the broader Andes, Cerrado and Pantanal regions, identified as having significant biogeographical characteristics unique and relevant to Amazonia (RAISG 2024). It encapsulates almost 850 million hectares (8,470,209km²) of territory and more than 350 million inhabitants across nine national boundaries, approximately 10 and 11 percent of which is accounted for by Peru, respectively: 96.6m hectares and 4.1m people (‘Amazonia Under Pressure 2020’ 2025, tbl. 1,2).

4.1.4 Policy Data and Interventions

We focus primarily on two Alternative Development and Forced Eradication policies: PIRDAIS and CORAH, respectively. ‘Alternative Development’ is identified by the year-level spending under the PIRDAIS programme, as implemented within the ‘Budget for Results’ (PpR) public policy framework (Resolución directoral N.º 024-2016-EF/50.01 2016). Within the PpR, Budget Programmes (PP), such as PIRDAIS, are delivered through the provision of a defined and measurable set of goods and services, or *actividades*. This can either be through specific initiatives such as crop substitution programmes and training programmes, or broader transfers and investments (Yupanqui Acosta 2020, 15–17).⁹ We consider the sum of all district-level transfers jointly, given the breadth of spending purposes covered throughout the period. ‘Forced Eradication’ refers to the number of parcels of agricultural land within which the CORAH programme conducted coca crop eradication. This data is self-reported by the programme, typically one year after all operations have been carried out, aggregated by district.

We match these measures of drug control intervention, at the district-level, to the coca-cultivation activity observed within the individual territories that constitute them. A spatial weighting transformation is subsequently applied, using the approach specified in 4.2.2.

4.1.5. Politics of the Peasantry and Resource Extraction

⁸ The Amazon Basin is commonly taken to describe the smaller subset of territories (692, 591 hectares), covering the Amazonas, Araguaia-Tocantins and Marajo Rivers (‘3. Amazonia | Atlas RAISG 2020’, n.d.).

⁹ PIRDAIS specifically corresponds to budget code 0072 (DEVIDA 2025).

Politics of the Peasantry

Drawing on the work of (Bebbington et al. 2008) and the Peruvian Social Studies Centre (CEPES), we consider historic records of past coca movements and union membership as useful proxies for communities' organisational capacity and their bargaining power in national drug policy. Bebbington identifies coca community mobilisation in the early 2000s as a key Peruvian social movement, alongside agrarian, feminist and indigenous activism. Because social movements arise from sustained collective action, taken in response to perceived injustice (Bebbington et al. 2008), we take such cases as evidence of coca-growers' ability to influence, resist and bargain, as theorised in peasantry politics (Goodhand, Ballvé, et al. 2024).

We use contemporary CONPACCP union membership estimates from the national 'Marches for sacrifice' protests of 2003–2004 (Allison Spedding Pallet and Hugo Cabieses Cubas 2004), noting CONPACCP remains Peru's largest and most established coca union (Mirella Elisabeth Henriette van Dun 2009). These data are explored for two purposes: to construct a 'union presence' metric (membership per cultivated hectare), and to identify 'union territories', more generally, where organisation is more prominent.

Cocalero Valley	CONPACCP Membership
Apurímac-Ene	11000
Alto Huallaga-Uchiza	2000
Alto Huallaga-Puerto Pizana	1500
Alto Huallaga-Aucayacu	3000
Alto Huallaga-Tingo María	3000
Aguaytía	1200
Pichis-Palcazu	1000
Monzón	2500
Jaén San Ignacio	260
TOTAL	25460

Table 1: Concentrations of Coca Union Membership during the Marcha de Sacrificio protests (2003-4)
Taken from Hugo Cabieses' estimates (Allison Spedding Pallet and Hugo Cabieses Cubas 2004, 17)

Extractive Imperatives

Following (Boone 2003), we capture extractive imperatives shaped by the Amazon's natural resource context, including analysing overlaps with resource concession blocks: land officially recognised by the Peruvian Government for potential resource wealth, but where rights have not yet been transferred or exercised. These areas, demarcated by the Ministry of Energy and Mines (MINEM) and the government (Francisco Tong Gonzáles and Fernando Montero

Alvarado 2009), are supplemented by another novel source of remote sensing data, showing observed mining activity from 2019-2023 (MapBiomass 2025b).

Our analysis in particular considers areas with unrealised extractive potential, limiting confounding from realised extraction, which provides local jobs and significant tax revenues via the *canon minero*—a framework taxing 50% of extractive revenues for local development and social spending (Bancalari and Rud 2024; MEF 2025).¹⁰

Extractive imperative, in these realised instances, could feasibly shape (increase) the likelihood of state intervention as well as (increase) subsequent livelihood outcomes, harming the interpretability of the underlying drug control policy strategy.

¹⁰ For some sense of scale, recent estimates placed *canon minero* revenues at over 25% of extractive district municipalities' annual budgets. (Bancalari and Rud 2024).

4.2 Empirical Strategy

To explore the role of specific state drug control policies in shaping the livelihood trajectories of coca crop producers, we propose a dynamic panel approach that attempts to address several methodological challenges, namely unobserved heterogeneity across coca-growing territories, potential endogeneity of policy implementation, and measurement error related to the spatial allocation of district-level policies to finer-grained, territory-level crop cultivation observations.

The proposed empirical strategy builds upon the ‘difference GMM estimator’ (Arellano and Bond 1991), attempting to identify causal effects of both coercive (forced eradication) and inclusive (Alternative Development) drug control strategies, whilst accounting for the persistent nature of coca cultivation.¹¹

4.2.1 Identification Strategy:

Our baseline econometric specification reflects a dynamic panel, as follows:

$$cocc_{it} = \alpha + \beta_1 cocc_{it-1} + \beta_2 CORAH_{it} + \beta_3 PIRDAIS_{it} + X'_{dt}\theta + \gamma_i + \varepsilon_{it}$$

Where: $cocc_{it}$ represents the total hectares of coca bush under cultivation in territory, i , at time t ; $CORAH_{it}$ and $PIRDAIS_{it}$ capture the exposure to forced eradication measures and Alternative Development assistance provided, respectively; X'_{dt} represents a vector of various district-level control variables; γ_i stands for territory-specific, time invariant fixed effects, such as conditions of the soil and climate, or historic cultural associations with the coca crop, and ε_{it} some ideosyncratic error.

Contrary to the assumptions of a classic static panel model, we allow for some temporal dependency in our outcome of interest through the inclusion of our lagged dependent variable $cocc_{it-1}$. We do so on the expectation that levels of coca cultivation at any given time will be, in some significant part, determined by previous levels. This reflects the frictions associated with exiting current markets and seeking new livelihood strategies, including considerations of

¹¹ We conduct our analysis in R, using the pgmm package (McNicholas et al. 2025).

upfront costs already incurred by (particularly land-owning) illicit peasantries, the slow growth in sources of alternative (legal) livelihood opportunities within many remote frontier territories (Thomson et al. 2022; Grillo 2018), as well as the time taken for coca crops to grow after they have first been planted.¹²

Addressing Unobserved Heterogeneity and Endogeneity Concerns

The presence of territory-level fixed effects, γ_i , presents a particular challenge for consistent estimation in the context of drug control policy.

Whilst factors correlated with drug crop cultivation dynamics, such as regional farmgate prices or unemployment rates (V. Greenfield et al. 2015; Rincón Ruiz et al. 2013), are observable and therefore possible to control for, others are not. Less tangible dimensions, such as persistent institutional factors or historic under-delivery of public services (Goodhand 2005), are often associated with both the conditions that drive communities to illicit drug crops and, later, the conditions that warrant state-building projects and (re)assertions of control (Paredes and Manrique 2021). We attempt to capture this unobserved heterogeneity, at least initially, through the use of the classic first-difference (FD) transformation:

$$\Delta cocc_{it} = \beta_1 \Delta cocc_{it-1} + \beta_2 \Delta CORAH_{it} + \beta_3 \Delta PIRDAIS_{it} + \Delta X'_{dt} \theta + \Delta \varepsilon_{it}$$

Where Δ stands for the change in the value, relative to the previous period (e.g. $\Delta Y_i = Y_{it} - Y_{it-1}$) and the unobservable variable(s), or previously ‘fixed effect’, γ_i , has been ‘differenced away’ (Angrist and Pischke 2008, 183).

A well-documented concern of the FD transformation in such applications, however, is its mechanical introduction of first-order autocorrelation (Kripfganz and Schwarz 2015; Olajide and Olubusoye 2015), via the presence of $cocc_{it-1}$ in both $\Delta cocc_{it-1}$ and the error term, $\Delta \varepsilon_{it}$. We attempt to mitigate this, alongside other unaddressed risks of endogeneity that arise from instances of reverse causality or simultaneity with regards to our policy variables,¹³ through employment of the Arellano-Bond (AB) GMM (General Method of Moments) estimator, in which lagged levels of coca cultivation are used as instruments for the first-

¹² Coca crops generally reach maturity after a year, after which they can provide a harvest every three-to-four months (Bradley and Millington 2008; Farthing and Grisaffi 2023, 4).

differenced equation, and policy variables are taken as weakly exogenous (Arellano and Bond 1991). Our proposed specification favours a limited number of lags to mitigate the risk of overfitting documented by (Roodman 2009), alongside similar concerns (Windmeijer 2005):

This relies on numerous identifying assumptions. First, that there is no serial correlation in idiosyncratic errors, such that $E[\varepsilon_{it}\varepsilon_{it-2}] = 0$. In other words, that no random variation in coca cultivation in year t is correlated with random variation in year $t - 2$. Diagnostic Arellano-Bond AR(2) tests,¹⁴ reported alongside our analyses, support this assumption.

Second, instrument exogeneity is assumed, such that past coca cultivation levels, at $t - 2$, are not correlated with random variation in cultivation in the present year, t . Whilst it is perhaps plausible that producers in a territory may anticipate policy interventions in the following period, for example, predictions two years in advance would appear less so. In any case, as in (Arellano and Bond 1991), Sargan-Hansen tests of overidentifying restrictions are provided to support this assumption.¹⁵

Third, it is necessary that the instrument, levels of coca cultivation two periods prior, $cocd_{it-2}$, influence current levels only via $cocd_{it-1}$ (the exclusion restriction). On the one hand, policy targeting based on historical coca levels would appear to represent a credible threat to this assumption. This threat is particularly relevant for levels observed in the immediately preceding period, given that targeting locations in Peru are typically identified and approved by the Presidency of the Council of Ministers (PCM)¹⁶ for the coming year (DEVIDA 2023b).

We take several steps to address this, including allowing for the possibility of heterogeneous treatment effects in territories that begin the period with higher levels of observed cultivation. We also present multiple model specifications, including those with extended lags that are less likely to be correlated with current policy targeting, alongside our existing Sargan test statistics.

The validity of the GMM estimator also rests upon and supports an assumption of weak exogeneity, such that our polices variable of interests, in a given period, are uncorrelated with

¹⁴ The AR(2) tests the following null hypothesis: $H_0: E[\Delta\varepsilon_{it}\Delta\varepsilon_{it-2}] = 0$.

¹⁵ The Sargan-Hansen test in IV-GMM estimation tests against the null hypothesis that all instruments are uncorrelated with ε , by regressing the residuals from an Instrumental Variable on the set of all instruments, Z .

¹⁶ *Presidencia del Consejo de Ministros*

current or past values, such that $E[PIRDAIS_{is}|\varepsilon_{it}] = 0$ for all $s \leq t$ (Wooldridge 2010). Under this assumption, lagged values of the explanatory variables serve as valid instruments, ensuring consistent estimation of the causal effect of eradication and alternative development policies. Insignificant results from a series of ‘placebo’, or ‘anticipation’ tests (See: **Appendix: Policy Anticipation Test**) in which potential effects of future policies, as ‘lead policy variables’, on cultivation are evaluated, lend support in this regard.

4.2.2 Spatial Weighting of Territorial Interventions

A final empirical challenge we note relates to measurement error. Whilst the SISCOD and MapBiomass datasets provide observations of cultivation and mining activity at a much finer spatial grain, within 10km x 10km grid cells, i , policy intervention data is generally only available at the district-level, d . A classical approach to this ‘map comparison’ problem (Jacquez 1995), to reconcile these different levels of aggregation, would be to interpret the district-level policy, be it $CORAH_{dt}$ or $PIRDAIS_{dt}$, as the average policy exposure for the subset of territories assigned to that district, where assignment is based upon the district with which the largest area of overlap is shared. That is, where $D = \{d_1, d_2, \dots, d_K\}$ represents the set of all Peruvian administrative districts, and $G = \{g_1, g_2, \dots, g_N\}$ represents the set of all grid cells, for each grid cell $i \in G$, assignment $d(i) \in D$ is determined by the maximum area of overlap.

In several cases, however, particularly where territories straddle multiple district boundaries, this area of overlap may vary significantly, and feasibly undermine an ‘average treatment’ interpretation. Similar to approaches adopted in epidemiology (Chambliss et al. 2024), we attempt to account for differing degrees of overlap within districts, and therefore potential variation in policy exposure, by predicting territory-level interventions based on the area-weighted averaging of this overlap, such that:

$$AreaWeight_i = \frac{Area(g_i \cap d(i))}{\sum_{j \in S_{d(i)}} Area(g_j \cap d(i))}$$

Where $Area(g_i \cap d(i))$ describes the geographic area of overlap between a grid cell, i , and its assigned district, $d(i)$, and $S_{d(i)} = \{j \in G : d(j) = d(i)\}$ denotes the subset of grid cells

assigned to district $d(i)$. From here, area-weighted policy variables are constructed as follows, in the case of PIRDAIS:

$$\widehat{PIRDAIS}_{it} = PIRDAIS_{dt} \times AreaWeight_i$$

This approach better reflects the role of administrative boundaries in shaping policy implementation than a naïve average treatment interpretation of arbitrary grid cell division would allow. It is also associated with greater instrument validity, as reflected in our initial Sargan test results.

We therefore estimate the following first-differenced dynamic panel using GMM, as our final specification:

$$\Delta cocc_{it} = \beta_1 \Delta cocc_{it-1} + \beta_2 \Delta \widehat{CORA}H_{it} + \beta_3 \Delta \widehat{PIRDAIS}_{it} + \Delta X'_{dt} \theta + \Delta \varepsilon_{it}$$

Where the lagged dependent variable $\Delta cocc_{it-1}$ is instrumented by $cocc_{it-2}$.

5. Analysis and Results

H1a: Coercive drug control policies will be associated with significant reductions in coca crop cultivation.

Table H1a: Dynamic Panel GMM Results: CORAH Eradication Effects

	<i>Dependent variable: Change in Coca Cultivation (hectares)</i>
Lagged Coca Cultivation	0.7747*** (0.1295)
CORAH Eradication (parcels)	-0.0035** (0.0011)
Model Diagnostics	
Sargan test p-value	0.336
AR(1) test p-value	0.008
AR(2) test p-value	0.743
Wald test p-value	< 0.001
Sample Information	
Territories	612
Observations	1,805
Method	One-step Difference GMM
Transformation	First differences
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 Standard errors robust to heteroskedasticity. Sample restricted to territories with initial coca cultivation.	

Our first specification provides modest support for Hypothesis 1a, with a statistically significant reduction ($P < 0.00182$) in total area under coca cultivation associated with CORAH eradication campaigns in a given territory, i . Each of our specifications also provide strong support for the ‘sticky’ logic behind our dynamic panel specification.

We may broadly interpret our most simple model, instrumenting $cocd_{it-2}$ for $\Delta cocd_{it-1}$, and accounting for the role of ‘persistence’ in determining levels of cultivation observed during the current period, t , with the coefficient on our instrumented $\Delta cocd_{it-1}$ attributing approximately

78% of year-on-year cultivation to levels recorded in the previous year. Additional variation in observed coca, at least in marginal terms, at the level of individual territories (hectares removed per parcel of farmland intervened upon by CORAH), variation induced by $CORAH_{it}$ might appear modest ($\beta = 0.0035$). Across the sum of all parcels of coca farmland targeted across the period ($N=217,919$), however, this scales to more significant levels (an estimated ~ 762 ha).

As we have noted previously, this variation is assumed to be induced by an exogenous regressor. We attempt to further substantiate this assumption by accounting for the possibility of heterogeneous treatment effects, such as in the possibility where elevated drug crop production invites unique state attention, in a way not already accounted for.

We do so through consideration of alternative specifications that allow for a differential eradication effect, by way of any interaction term on $CORAH_{it}, HighExistingActivity_i$, defined as the set of territories in the top 25th percentile of cultivation sites at the beginning of the period in 2019. We also include the same model with deeper lags, to address similar concerns discussed previously. Whilst a significant, positive differential is observed in both cases, consistent with other observations, it is noteworthy that both the validity of our instrument, and the significance of our $CORAH_{it}$ effect remain broadly robust to these considerations.

We note finally, that in all cases, AR(1) diagnostic tests confirm the presence of first-order correlation within the First-Differenced errors, as expected by construction, whilst AR(2) tests provide no evidence of second-order autocorrelation, as required by our identifying assumptions. P values for all Sargan tests of overidentification in model instruments are also above conventional thresholds of statistical significance (Labra Lillo and Torrecillas 2018).

H1b: Inclusive drug control strategies will have a weak negative relationship with coca crop cultivation, at least in the short term.

By its nature, we would expect instances in which ‘the governing’ (the state) are either less willing, or less capable, of unilaterally imposing their will on the ‘governed’, that those interests (here, including more traditional counter-narcotics objectives), will be realised less expediently. That is to say that more inclusive territorial governance regimes, even where still directed at overall reductions in coca crop cultivation, may be associated with much less prominent shifts.

Indeed, under our most simple model specification, instrumented by a single lagged instrument for coca persistence and no additional controls, $PIRDAIS_{it}$ drug control strategies appear to lack the same evidence of statistical significance as was demonstrated by more coercive approaches (see **Table H1a** above).

As reported in several territories (Grillo et al. 2021), we consider the possibility of incomparable legal market incentives within AD programmes, and of shifting coca-cocaine market dynamics as a potential source of spatial and temporal variation in $PIRDAIS_{it}$ impacts.

Official coca leaf pricing data, available at the Department-level (SISCOD 2024) , fails to capture sufficient variation at the territory-level to offer useful analysis. We therefore consider a reduced sample of territories for which more localised, price data are available to explore this, within more regularly monitored ‘coca-growing zones’.¹⁷

Whilst integration of the price trends at different stages of the coca-cocaine supply chain would initially appear to present the $PIRDAIS_{it}$ negative effects as statistically significant (*See **Model 1 Table H1b***), the corresponding, negative values on relevant market coefficients would appear to have little grounding in empirical reality. We therefore treat these results with suspicion, perhaps as the result of multicollinearity (Kim 2019) between PIRDAIS’ market intervention instruments and related dynamics within the coca-cocaine supply chains being targeted.

¹⁷ We specifically look at price monitoring from cocalero ‘subzones’, defined by DEVIDA (DEVIDA and OPD, n.d.; DEVIDA 2024c; OPD 2023).

Table H1b: Price Dynamics and Policy: Specification Sensitivity and Union Moderation

		<i>Dependent variable: Change in Coca Cultivation (hectares)</i>	
		Model 1	Model 2
		<i>Main Effects Only</i>	<i>With Union Interaction</i>
	Lagged Coca Cultivation	0.8507*** (0.1282)	0.8508*** (0.1282)
	PIRDAIS spending per Rural Family	-0.0001*** (0.0000)	-0.0000 (0.0002)
	PIRDAIS × Union Density	—	-0.0001 (0.0002)
Market Controls	Local Coca Leaf Price	-0.6386† (0.3507)	-0.6360† (0.3512)
	Local Cocaine Price	-0.0022 (0.0015)	-0.0022 (0.0016)
Model Diagnostics			
	Sargan test p-value	0.213	0.206
	AR(1) test p-value	0.003	0.003
	AR(2) test p-value	0.440	0.439
	Wald test p-value	< 0.001	< 0.001
Sample Information			
	Territories	526	526
	Observations	1,550	1,550
	Method	One-step Difference GMM	One-step Difference GMM
<i>Note:</i>		***p<0.01; **p<0.05; *p<0.1; †p<0.15	
		Standard errors robust to heteroskedasticity.	

A likely confounder within such regions may lie in their historic connections to coca unions, or *sindicatos cocaleros*, which often developed around many of the same landscapes of coca dependence and resistance that led the Peruvian state to take greater interest in monitoring and otherwise intervening.

As we have explained, the nature of territorial governance dynamics, and subsequently livelihood trajectories, are not just a reflection of ‘the governing elite’ and their capacities, but also of ‘the governed’, and their bargaining power. We therefore consider whether spaces of overlap with historic strongholds of organised cocalero activity, as in the valleys of the Upper Huallaga, Apurímac river and the Southern Amazon with the CONPACCP during the 1990s and early 2000s (Allison Spedding Pallet and Hugo Cabieses Cubas 2004), may provide a better conduit for collective interests to be realised.

Very weak evidence from this proxy measure (*See Model 2 Table H1b*) would suggest that coca union membership *itself*, or legacies of it, cannot guarantee effective bargaining with the state. Indeed, those like (Bebbington et al. 2008) highlight the ideological radicalism that many cocalero social movements have been borne from, and which has increasingly seen displaced peasant, *campesino*, communities and lowland indigenous groups in the region mobilise against the political elite and ever-more concentrated extractive economies (Bebbington 2018). Such attitudes may not always be conducive to immediately productive processes of bargaining.

2. Under what conditions do households move away from drug crop livelihoods as part of an inclusive transition?

Subsequent analysis attempts to reframe how the impacts and broader role of recent drug control policy have shaped coca-growing communities' material conditions in parts of the Peruvian Amazon, with greater emphasis on human-centred development outcomes and sustainable livelihood outcomes.

Following a similar methodological approach to (Rozo 2014), who evaluates the welfare implications for drug control policy in Colombia, we first propose a small adaptation to the previous empirical specification. To estimate the corresponding causal effects on relevant, alternative sources of economic opportunity, $AltLivelihoods_{it}$, that results from reductions in coca crop cultivation, we evaluate the following model:

$$\Delta AltLivelihoods_{it} = \phi AltLivelihoods_{it-1} + \theta \Delta cocc_{it-1} + \beta X'_{it} + \mu_i + \varepsilon_{it}$$

But this time using observed change in coca cultivation as an explanatory variable, instrumenting $cocc_{it-2}$ for $\Delta cocc_{it-1}$, whilst capturing the similar persistence, ϕ , in relevant legal agricultural livelihood strategies, $AltLivelihoods$, as before with coca cultivation.

H2a: Reductions in coca cultivation, associated with the introduction of Coercive drug control strategies, will have no significant relationship with any associated changes in Livelihood Outcomes.

Perhaps contrary to initial predictions under H2a, our results provide tentative evidence of a 'crowding out' effect of legal crops, $AltLivelihoods$ here representing the sum of all rice plantations and other major perennial crops, as tracked by the remote sensing organisation (MapBiomas 2025b).

Table H2a: Coca Cultivation Effects on Legal Crop Livelihoods

<i>Dependent variable: Change in Legal Crop Cultivation (hectares)</i>		
	Model 1	Model 2
	<i>Instruments: lag(2)</i>	<i>Instruments: lag(2:3)</i>
Lagged Legal Crop Cultivation	0.7968*** (0.1291)	0.7400*** (0.0881)
Lagged Coca Cultivation Change	-0.0055** (0.0025)	-0.0049** (0.0023)
CORAH Eradication	-0.0011 (0.0008)	-0.0011 (0.0008)
Model Diagnostics		
Sargan test p-value	0.019	0.111
AR(1) test p-value	0.011	0.008
AR(2) test p-value	0.741	0.693
Wald test p-value	< 0.001	< 0.001
Sample Information		
Territories	612	612
Observations	1,802	1,802
Method	One-step Difference GMM	One-step Difference GMM

Note:

***p<0.01; **p<0.05; *p<0.1

Standard errors robust to heteroskedasticity.

Whilst our first model, instrumented by only a single period lag, appears to raise overidentification concerns, as per the Sargan test result ($P < 0.02$), our second model would appear to carry some statistically note-worthy implications: that a 1 hectare reduction in coca cultivation (that is, as an indirect consequence of CORAH eradication intervention), during the previous period, $t - 1$, is associated with a corresponding increase in overall legal crop activity of 0.0049 hectares, during the current year, t . This would, on the one hand, appear a classical manifestation of the ‘crowding out’ effect appealed to within more mainstream drug control discourse, bringing greater private investment and economic opportunity to former ‘narco-frontiers’.

We emphasise, however, that at this level of aggregation, it remains unclear who the true ‘winners and losers’ are of the resulting agricultural transition, and of the precise nature of the livelihood arrangements being observed.

H2b: The strongest corresponding increase in Livelihood Outcomes will be associated with reductions in coca cultivation that are associated with: the implementation of Inclusive drug control strategies; the presence of nearby extractive imperatives; and the existence of strong bargaining power on the part of coca growers.

Like the previous hypothesis, we also considered any associated livelihood changes resulting from coca cultivation changes, where those changes were induced by Alternative Development interventions, *PIRDAIS*.

It should be made clear from previous analysis, however, that *PIRDAIS*-related spending has itself not been consistently associated with robust reductions in coca. Analysis of livelihood outcomes that are associated with such reductions is therefore unlikely to be robust, similarly.

Indeed, we find little conclusive evidence from considering two potential sites of livelihood activity: legal agricultural activity, on aggregate (as before, with *CORAH*), as well as observed activity in markets for cash crops, an area that is theoretically more directly tied to *PIRDAIS* substitution initiatives.

In the first model (overall agricultural activity), a severe Sargan test result allows us to confidently reject the Null hypothesis that all overidentifying restrictions valid, and thus that are instruments can provide valid causal interpretation. Whilst the second model (focused specifically on commodity crop activity) may at least provide more valid instruments, *AR(2)* diagnostic tests results such that second-order serial correlation are a threat to more serious consideration.

Table H2b: Mining Presence and Development Policy: Differential Effects on Agricultural Outcomes

	<i>Dynamic Panel GMM Estimation (One-Step, First Differences)</i>	
	Model 1	Model 2
	<i>Total Agricultural Area</i>	<i>Cash Crop Cultivation</i>
Lagged Legal Crop Cultivation	0.1483*	-0.1628
	(0.0722)	(0.1070)
Lagged Coca Cultivation Change	0.0050*	-0.1831
	(0.0021)	(0.1053)
PIRDAIS spending per Rural Family	-3.201e-07	-3.613e-05*
	(1.719e-07)	(1.842e-05)
PIRDAIS × Mining Area	-1.385e-05	-7.319e-03
	(7.130e-06)	(5.970e-03)
Model Diagnostics		
Sargan test p-value	4.74e-06	0.147
AR(1) test p-value	0.106	0.017
AR(2) test p-value	0.383	4.45e-05
Wald test p-value	0.006	< 0.001
Sample Information		
Territories	526	526
Observations	1,542	1,570
Instrument	lag(coca, 2)	lag(coca, 2)

Note: ***p<0.01; **p<0.05; *p<0.1; p<0.1

Standard errors robust to heteroskedasticity.

It is telling that in both models, coefficients on PIRDAIS spending is extraordinarily small (-3.201e-07, and -3.613e-05, respectively), perhaps further highlighting the differences in scale at which Alternative Development approaches have been implemented, by contrast with historically dominant enforcement-led and forced eradication approaches (à la CORAH). Before we touch on this point further, however, we note something in brief:

Whilst not a sufficient basis for causal inference, a higher-level view of within-district correlation between year-on-year PIRDAIS spending and agricultural activity is suggestive that interventions are frequently associated with some growth in alternative legal cash crop industries. A time series cross correlation of annual PIRDAIS investment and reported cultivation in cacao, coffee and plantain reveals some immediate, positive correlation ($P < 0.095$) between intervention and cultivation in 4 districts (see Figure below), with Pearson's correlation coefficients given below. Though PIRDAIS has not necessarily driven progress in livelihood outcomes through coca reductions, there is perhaps at least tentative evidence to suggest it has begun to improve them *alongside* coca.

District of Intervention	Inclusive Intervention (with relevant lag)	Alternative Livelihood Measure	Pearson's Correlation Coefficient
CHOLON	$TotInvest_{PIRDAIS\ t_0}$	$Commodity_{Crop_{t_0}}$	0.989***
CODO DEL POZUZO	$TotInvest_{PIRDAIS\ t_0}$	$Commodity_{Crop_{t_0}}$	0.928**
SAN PEDRO DE PUTINA PUNCO	$TotInvest_{PIRDAIS\ t_0}$	$Commodity_{Crop_{t_0}}$	0.892**
VILLA VIRGEN	$TotInvest_{PIRDAIS\ t_0}$	$Commodity_{Crop_{t_0}}$	1.000***
CONSTITUCION	$TotInvest_{PIRDAIS\ t_{-2}}$	$Commodity_{Crop_{t_0}}$	0.999**
MONZON	$TotInvest_{PIRDAIS\ t_{-2}}$	$Commodity_{Crop_{t_0}}$	-0.997**
VILLA KINTIARINA	$TotInvest_{PIRDAIS\ t_{-2}}$	$Commodity_{Crop_{t_0}}$	1.000***

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$; $p < 0.1$

Table H2b: Correlation Table of District-Level PIRDAIS Spending and Reported

Revealingly, Alternative Development's relationship with these livelihood outcomes does not appear to be stable across space, or *time*, with 2 additional districts only seeing significant correlations emerge ($P < 0.95$) when a lag of 2 years, from the point of PIRDAIS intervention, is accounted for. Even then, the appearance of a strong negative correlation with future cultivation, in the district of Monzón, leaves several questions unanswered.

Such findings suggest further localisation is needed to understand more complex interactions between territorial dynamics and state drug control regimes.

6. Discussion and Conclusion

Data scientists and economists alike have described “the flattening of the world” under the era of ‘big data’ (M.K. Rhemann 2017), as sources of satellite imagery become ever-more available, and ever more information can be obtained about the previously inaccessible. That being the case, blind trust in such technologies and the ‘quantification’ of realities, perhaps equally risks a “flattening of the debate” (IPDC 2022, 10), and the opportunity to miss important nuances at play in the complex, territorial dynamics of narco-frontiers.

The present work makes several trade-offs by design, not least to balance the breadth of scope demanded by a study of frontiers from a wider, regional perspective, with a recognition of the need for more localised units of analysis, consistent with the ‘spatial unevenness’ of socio-territorial dynamics (Paredes et al. 2024). On the one hand, the Difference GMM specification proposed here smooths much of the ‘noise’ in the vastness of data considered to identify some noteworthy trends across territories.

Perhaps most clearly, our results illustrate the ‘stickiness’ of coca crop cultivation as a form of livelihood strategy. Across most of our results, we find that anywhere from 75% to 80%+ of coca crop activity in a given year can be predicted from levels observed in the previous year. This resonates with the discourses in the drug policy literature, particularly the “dependence” of frontier communities on drug crops (V. Greenfield et al. 2015, 9). Further, in considering potential interactions with local actors, interests, and the politics of coca unions, we find that their roles and how they will be manifested in resulting bargains cannot be easily generalised or predicted. The coca movement and its members have been described as unique from other social movements in many ways (Bebbington et al. 2008), and this fact deserves more focused attention.

On the other hand, it seems evident that many of the nonstationary dynamics, including the particularities of PIRDAIS’ influence on local livelihood outcomes, cannot be effectively captured by the more restricted, linear assumptions of the approach posed thus far. Weak statistical power and small effect sizes in many cases suggest that the scale of interventions provided has been insufficient to see coca-growing communities make a transition away from dependency on coca crop livelihoods, with illicit markets seeming to prevail. Where there is a

more tentative, descriptive basis for optimism, in its association with growing legal markets in substitute cash crops, as with cacao, coffee and plantain, the delays associated with any longer-term results are varied, and more research is required to understand them further. Relatedly, the constrained period within which current coca crop data is available, alongside a lack of other, more localised contextual data, would appear to largely limit much analysis to ‘short-run’ interactions with coca-growing communities, despite more inclusive drug control strategies likely needing more time before many of their results can be observed.

Whilst more coercive state strategies may be able to demonstrate an immediate and tangible response to these sites, in the form of short-term forced eradications, there are reasons to be concerned about the extent to which this will provide longer-term answers to the conditions that constructed those initial dependencies.

Whilst our evidence suggests that CORAH-induced coca crop reductions may be associated with a marginal increase in some common licit crops that can be classified by satellite imagery (MapBiomass 2025a), this does not guarantee that the proceeds of this newly-created licit market activity will be enjoyed by those same, former coca crop growers. In fact, market growth of the speed and scale most likely to be captured by these methods, and over a relatively short period of observation, is perhaps more characteristic of capital-intensive, agro-industrial expansion, than gradual piecemeal adaptation on the part of individual farmers and growing communities. Such dynamics may echo aspects of recent work by Ballvé, who has highlighted ‘narco-frontiers’ as providing the ideal conditions for runaway agribusiness and unregulated ‘narco-capital’ to bring about widespread dispossession of coca-growing communities (Ballvé 2019; Ballvé and McSweeney 2020).

As well as its many shortcomings, the present study makes clear the many opportunities that such approaches might offer for future research in licit and otherwise less accessible contexts. The key to their successful use will undoubtedly be in ensuring they are matched with the insight and interdisciplinary expertise required to separate the signal from the noise, without losing too much of what is important.

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Appendix

Appendix: Policy Anticipation Test: Robustness Check

	<i>Dependent variable: Change in Coca Cultivation (hectares)</i>	
	Placebo Model 1	Placebo Model 2
	<i>Instruments: lag(coca, 2)</i>	<i>Instruments: lag(coca, 2:3)</i>
Lagged Coca Cultivation	0.7727*** (0.1289)	0.7450*** (0.1279)
Future CORAH (t+1, z-score)	-0.606 (2.702)	-0.444 (2.637)
Future PIRDAIS (t+1, z-score)	1.037 (3.418)	1.027 (3.400)
Model Diagnostics		
Sargan test p-value	0.325	0.145
AR(1) test p-value	0.008	0.010
AR(2) test p-value	0.749	0.752
Wald test p-value	< 0.001	< 0.001
Sample Information		
Territories (n)	612	612
Observations Used	1,810	1,810

Note: *** p<0.01; ** p<0.05; * p<0.1

Standard errors robust to heteroskedasticity.

Year	National Programme	Subject of Intervention	Relevant Aspects	Notable Criticisms
1982 - Present	The Special Project of Control of Coca Crops in the Upper Huallaga (CORAH) <i>Proyecto Especial de Control de Cultivos de Coca en el Alto Huallaga</i>	Forced eradication of coca crops in identified <i>cocalero</i> zones. From 1994, the scope of intervention has been the entire country.	Administered by the Peruvian Ministry of the Interior (MININTER).	Inflammation of state-society tensions; Failure to address underlying conditions driving coca cultivation; Worsening livelihood outcomes
1982 - 1994	The Plan for National Rehabilitation (PNR) <i>Plan Nacional de Rehabilitación</i>	Conceived as ‘physical crop substitution’ programme in post-eradication <i>cocalero</i> zones. Intended to help CORAH mitigate community discontent and disruption caused to livelihoods.	Implemented by CORAH, with financial support by the U.S.	Delivery hampered by poor state-society relations; Failed to address broader challenges relating to market access and uncompetitive legal crop prices; Technical failures, such as mismatched crops for local soil and climate conditions.
1995	The Alternative Development Programme (PDA) <i>Programa de Desarrollo Alternativo</i>	Follows Integrated Rural Development (IRD) approach. Interventions included investment in greater productivity within legal supply chains. Community and farmer support for more efficient agricultural and agroindustrial marketing.	Primarily funded and implemented by USAID, with some assistance from local governments.	Prioritising of forced eradication strategies alongside the PDA undermined state-society relations. Failure to facilitate long-term transitions away from coca crop livelihoods.
2014 - Present	Implementation of the ‘Integrated and Effective Management of Drug Supply Control (GIECOD) <i>Gestión Integrada y Efectiva del Control de Oferta de Drogas</i>	Broader reinforcing of counter-narcotics initiatives across law enforcement agencies and the courts. Includes interdiction of illicit drug production and trafficking activity, alongside support for criminal persecution. Surging of resources for military presence in areas designated a ‘State of Emergency’.	Coordinated and funded through DEVIDA. Cross-agency collaboration with MININTER and DIRANDRO (Anti-Narcotics Unit of the Peruvian National Police).	Inadequate personnel training; Allegations of police corruption and lack of public trust; Poor resourcing and telecommunications infrastructure; Seizures of chemical inputs insufficient to reduce cocaine supply.
2014 - Present	Implementation of the ‘Comprehensive and Sustainable Alternative Development Budget Program’ (PIRDAIS)	Revised and expanded previous PDA approaches. Alternative livelihood support at the household-level. Regional-level investment in market access and transport infrastructure. Capacity-building initiatives at the community and local government-level.	Coordinated and funded through DEVIDA. Administered across national, regional and local-level government.	Development support conditional on full, voluntary eradication of existing coca stocks; Coinciding enforcement operations undermine state-society relations. Inequitable access to crop substitution and training programmes.

Appendix: The Evolution of Drug Control Policy in Peru

Source: Author's own elaboration from (Paredes et al. 2024; Kaulard 2018; Guerra 2014; Grillo et al. 2021)

Appendix: H1a Table: Potential Heterogeneity of Forced Eradication Treatment Effects by Initial Coca Activity: Robustness Across Instrument Specifications

	<i>Dependent variable: Change in Coca Cultivation (hectares)</i>	
	Model 1	Model 2
	<i>Instruments: lag(2:3)</i>	<i>Instruments: lag(2)</i>
Lagged Coca Cultivation	0.7539*** (0.1279)	0.7815*** (0.1289)
CORAH × Lower Initial Coca Activity	-0.0025*** (0.0007)	-0.0025*** (0.0070)
CORAH × High Initial Coca Activity	-0.0306** (0.0135)	-0.0294** (0.0135)
Model Diagnostics		
Sargan test p-value	0.134	0.347
AR(1) test p-value	0.009	0.008
AR(2) test p-value	0.738	0.734
Wald test p-value	< 0.001	< 0.001
Sample Information		
Territories	612	612
Observations	1,805	1,805
Method	One-step GMM	One-step GMM

Note:

*p<0.1; **p<0.05; ***p<0.01

High initial activity = top 25% of coca cultivation in 2019.

Standard errors robust to heteroskedasticity.

Type	Variable Name	Units	Description	Territories (N=612)		Source
				Mean	(Std Dev)	
Agricultural Livelihoods	Coca Cultivation	Hectares	Areas of land under coca bush cultivation, per annum, identified by DEVIDA's SISCOD drug crop monitoring programme.	106.11	(258.03)	(UNODC and DEVIDA 2025).
	Legal Crop Cultivation and Activity	Hectares	Areas of land usage classified by major crop type, including for rice, palm oil and other perennial crops, annually. Identified by MapBiomias.	56.94	(462.15)	(MapBiomias 2025a)
	Substitute Cash Crop Cultivation	Hectares	Cultivation areas of primary substitute cash crops used within PIRDAIS intervention sites, specifically cacao, coffee and plantain. Reported annually by the Department of Agriculture.	463.51	(776.58)	(MIDAGRI 2024)
Policy Variables of Interest	Forced Eradication (CORAH)	Parcels of land	The number of parcels of agricultural land intervened upon by drug enforcement agents, under the Ministry of the Interior's CORAH programme.	71.22	(448.61)	(CORAH 2024)
	Alternative Development (PIRDAIS)	Constant 2023 Peruvian Soles (/S) per Rural Family	Total annual transfers made, per rural family, at the district-level, under the PIRDAIS programme (Budget Code 0072).	668.28	(1095.59)	(DEVIDA 2024a; 2023a; 2022; 2020b; 2020a)
Politics of the Peasantry	Coca Union Presence	(Union Membership / Hectares Grown)	Based upon contemporary reporting of CONPACCP membership statistics and locations. Cross-referenced with DEVIDA-identified <i>zonas cocaleros</i> .	2.38	(1.03)	(Allison Spedding Pallet and Hugo Cabieses Cubas 2004; DEVIDA 2024b)
Extractive Imperative	Mining Locations	Hectares of Mining Activity	Areas with mining concessions in place, cross-referenced with confirmed sites of mining activity, identified by MapBiomias.	2.46	(18.92)	(MapBiomias 2025b)
Demographic Measures	Total Population	People	Taken from the national statistics office's (INEI) population projections, from the 2017 census.	1817	(3024.38)	(INEI 2025)
	Rural Families	(Annual Population / Average Rural Occupancy)	Calculated based on rural population statistics and district average household occupancy estimates, taken from the INEI's ENAHO household survey.	539.25	(909.84)	(INEI 2024)
Broader Market Demand	Wholesale coca leaf prices	Constant 2023 Peruvian /S	Average of annual prices paid in wholesale markets within coca-producing zones. Taken from annual reporting, conducted by DEVIDA, under their SISCOD platform.	9.34	(2.47)	(SISCOD 2024)
	Wholesale coca paste prices	Constant 2023 Peruvian /S		2817.27	(342.75)	
	Wholesale cocaine prices	Constant 2023 Peruvian /S		5256.67	(832.74)	

Appendix: Descriptive Statistics